

RAJ MAKWANA

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Experience

PayPal Inc.

May 2026 – August 2026

Software Engineer Intern

Bengaluru, Karnataka

- Engineered a self-service **data pipeline** for a compliance investigation platform, replacing manual onboarding and validation workflows with automated stages processing **billions of rows**.
- Designed **BigQuery tables** for auditability, historical baselines, data statistics, governance, and checkpoints, establishing traceability and system-of-record capabilities for data workflows.
- Migrated **graph computation** from BigQuery SQL to **Apache Spark/Scala**, optimizing for large-scale data volume through **salting, AQE, join reordering, broadcast joins, checkpointing, and cluster tuning**.
- Developed a **validation stage** with **native sql checks** covering data integrity, domain rules, and computed-result correctness.

Matterassist

January 2025 – December 2025

Full-Stack Engineer Intern

Ahmedabad, Gujarat

- Developed **RESTful APIs** with **Node.js**, utilized **MongoDB** for data persistence and Cloudinary for object store.
- Designed architecture of essential feature **Teams** from **end to end and scratch**, integrating modular components for team creation, team details-integrated **folder system** for team files with this, search member and request management.
- Developed the **Bill** module with dashboard, letter-pad bill form, and invoices; implemented event notifications via **React Toastify** and profile-view state sharing via **React Context API**.

Projects

Event Driven Ledger | *PostgreSQL, Redis, Java, SpringBoot, Kafka, Docker* | [Github](#)

- Architected and built a **distributed**, event-driven ledger system using **CQRS** and **microservices**, enabling strict separation of write and read models with **Kafka-based asynchronous** communication.
- Designed a **resilient** command processing pipeline with **idempotency**, **transactional outbox pattern**, and **optimistic concurrency** with internal retry mechanisms to guarantee **consistency** under concurrent workloads.
- Engineered **high-performance** write model using **snapshot replay optimization** and read models using **PostgreSQL** projections, indexing and **Redis** caching, significantly reducing **latency** and improving **scalability**.
- Delivered end-to-end **system design** including **HLD, LLD**, API contracts, and **UML** artifacts (component, class, ER, sequence diagrams), applying **SOLID** principles, load balancing, and fault-tolerant distributed system patterns.

ChatGram | *MongoDB, React Js, Node Js, SocketIO, Gemini API* | [Github](#)

- Designed **interrelated Database schemas** and developed APIs for authentication using **JWT** and bcryptJs, **CRUD** of modules including profile, message and friend requests.
- Integrated **SocketIO** for live message syncing and Gemini API for AI chat bot interaction.

Technical Skills

Languages: C++, Java, Python, JavaScript

Databases: MySQL, MongoDB, PostgreSQL

Frameworks/Technologies: Spring Boot, Apache Spark, Apache Kafka, MERN Stack, Docker, REST APIs

Concepts: Distributed Systems, Microservices Architecture, Event-Driven Architecture

Cloud: AWS, GCP, Vercel, Firebase

Achievements

- Bronze Medalist**, First Year Engineering – Awarded for outstanding academic performance
- Hacktoberfest** '24 and '25 – Earned all-level badges through **Open Source Contribution**
- Leetcode **Knight** (Top 3.94%, Max rating.1916) – Achieved Rank 672 in Weekly Contest 471, 25,000+ participants
- Codeforces **Pupil** (Max rating.1300) – Achieved Rank 3347 in Round 1032, 30,000+ participants ([Profile Link](#))
- Atcoder **Green** (Max rating.887) – Achieved Rank 1583 in Round 423, 10,000+ participants ([Profile Link](#))
- IICPC CodeFest'25 Prelims – Achieved Rank **1,942**, 45,000+ participants ([Certificate](#))
- Meta Hacker Cup 2025 – Secured Rank **5,080** in Round 1 and **3,272** in Round 2 ([Certificate](#))
- NASA Space Apps Challenge 2024** — **Team Lead** – Led the team to become a **Top 5 finalist** among **50+ teams** in the regional university competition.

Education

Nirma University

August 2023 – June 2027

Bachelor of Technology in Computer Science and Engineering : Grade – 9.17

Ahmedabad, Gujarat

Relevant Coursework:

Data Structures, OOP, Algorithm Analysis, Database Management, Operating Systems, Computer Networks
Cloud Computing, Machine Learning, Reinforcement Learning, Soft Computing, NLP, Graph Theory